Summary

Newton's laws of motion essentially define the means by which motion changes and the relationship between these changes in motion with force and mass.

Underlying knowledge and understanding

Learners should have an understanding of contact and non-contact forces influencing the motion of an object. They should be aware of the newton and that this is the unit of force. The three laws themselves will be new to the learners. Learners are expected to be able to use force arrows and have an understanding of balanced and unbalanced forces.

Common misconceptions

Learners commonly have misconceptions about objects needing a net force for them to continue to move steadily and can struggle to understand that stationar objects also have forces acting on them. Difficulties faced by learners when tryin to differentiate between scalar and vector quantities is the idea of objects with a changing direction not having a constant vector value, for example, objects moving in a circle. This issue also arises with the concept of momentum and changes in momentum of colliding objects.

Tiering

Statements shown in **bold** type will only be tested in the Higher Tier papers.
All other statements will be assessed in both Foundation and Higher Tier papers.

Reference	Mathematical learning outcomes	Mathematical skills
PM2.2i	recall and apply: force (N) = mass (kg) × acceleration (m/s^2)	M1a, M2a, M3a, M3b, M3c, M3d
PM2.2ii	**recall and apply: momentum (kg m/s) = mass (kg) × velocity (m/s)**	**M1a, M2a, M3a, M3b, M3c, M3d**
PM2.2iii	recall and apply: work done (J) = force (N) × distance (m) (along the line of action of the force)	M1a, M2a, M3a, M3b, M3c, M3d
PM2.2iv	recall and apply: power (W) = $\frac{\text{work done (J)}}{\text{time (s)}}$	M1a, M2a, M3a, M3b, M3c, M3d

Topic content			Opportunities to cover:		Practical suggestions
Learning outcomes		To include	Maths	Working scientifically	
P2.2a	recall examples of ways in which objects interact	electrostatics, gravity, magnetism and by contact (including normal contact force and friction)			
P2.2b	describe how such examples involve interactions between pairs of objects which produce a force on each object				
P2.2c	represent forces as vectors	drawing free body force diagrams to demonstrate understanding of forces acting as vectors	M5b	WS1.2a, WS1.2b, WS1.2c, WS1.2e, WS1.3a, WS1.3c, WS1.3e, WS1.3h, WS2a, WS2b, WS2d	Measurement of the velocity of ball bearings in glycerol at different temperatures or of differing sizes. (PAG P3)
P2.2d	apply Newton's first law to explain the motion of an object moving with uniform velocity and also an object where the speed and/or direction change	looking at forces on one body and resultant forces and their effects (qualitative only)		WS1.3e, WS2a	Demonstration of the behaviour of colliding gliders on a linear air track. (PAG P3) Use of balloon gliders to consider the effect of a force on a body.
P2.2e	**use vector diagrams to illustrate resolution of forces, a net force (resultant force), and equilibrium situations**	**scale drawings limited to parallel and perpendicular vectors only**	M4a, M5a, M5b		

P2.2f	**describe examples of the forces acting on an isolated solid object or system**	**examples of objects that reach terminal velocity for example skydivers and applying similar ideas to vehicles**		WS1.2a, WS1.2b, WS1.2c, WS1.2e, WS1.3a, WS1.3c, WS1.3e, WS1.3h, WS2a, WS2b, WS2d	Learners to design and build a parachute for a mass, and measure its terminal velocity as it is dropped. (PAG P3)
P2.2g	**describe, using free body diagrams, examples where two or more forces lead to a resultant force on an object**				
P2.2h	**describe, using free body diagrams, examples of the special case where forces balance to produce a resultant force of zero (qualitative only)**				
P2.2i	apply Newton's second law in calculations relating forces, masses and accelerations		M1a, M2a, M3b, M3c, M3d	WS1.2a, WS1.2b, WS1.2c, WS1.2e, WS1.3a, WS1.3c, WS1.3e, WS1.3h, WS2a, WS2b, WS2c, WS2d	Use of light gates, weights and trolleys to investigate the link between force and acceleration. (PAG P2)

P2.2j	**explain that inertia is a measure of how difficult it is to change the velocity of an object and that the inertial mass is defined as the ratio of force over acceleration**				
P2.2k	**define momentum and describe examples of momentum in collisions**	**an idea of the law of conservation of momentum in collisions**		WS1.2a, WS1.2b, WS1.2c, WS1.2e, WS1.3a, WS1.3c, WS1.3e, WS1.3h, WS2a, WS2b, WS2c, WS2d	Use of light gates, weights and trolleys to measure momentum of colliding trollies. (PAG P3) Use of a water rocket to demonstrate that the explosion propels the water down with the same momentum as the rocket shoots up.
P2.2l ☑	apply formulae relating force, mass, velocity and acceleration to explain how the changes involved are inter-related		M3b, M3c, M3d		
P2.2m	use the relationship between work done, force, and distance moved along the line of action of the force and describe the energy transfer involved		M1a, M2a, M3a, M3b, M3c, M3d	WS1.4a, WS2a, WS2b	Measurement of work done by learners lifting weights or walking up stairs. (PAG P5)
P2.2n	calculate relevant values of stored energy and energy transfers; convert between newton-metres and joules		M1c, M3c	WS1.4e, WS1.4f	
P2.2o	explain, with reference to examples, the definition of power as the rate at which energy is transferred				
P2.2p	recall and apply Newton's third law	application to situations of equilibrium and non-equilibrium			
P2.2q	**explain why an object moving in a circle with a constant speed has a changing velocity (qualitative only)**			WS1.3e	Demonstration of spinning a rubber bung on a string.